|INCH-POUND|

MIL-E-80030C
12 September 1989
SUPERSEDING
MIL-E-80030B
2 July 1973

MILITARY SPECIFICATION
MILLING MACHINE, TWO AND THREE DIMENSIONAL ENGRAVING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two and three dimensional engraving machines, floor mounted and bench mounted, electric motor driven, for reproducing from masters; profile milling and engraving with the equipment and accessories specified herein.

1.2 Classification. The machines covered by this specification shall be of the following types, styles, and sizes. The type, style, and size supplied shall be as specified (see 3.5 and 6.2).

Type I - Bench Mounted
Style A - Two dimensional

Size 1 - 1/8 inch spindle capacity
Size 2 - 1/4 inch spindle capacity (1/15 HP)
Size 3 - 1/4 inch spindle capacity (1/6 HP)
Size 4 - 5/16 inch spindle capacity

Type II - Floor Mounted
Style A - Two dimensional

Size 1 - 1/4 inch spindle capacity
Size 2 - 5/16 inch spindle capacity
Size 3 - 3/8 inch spindle capacity
Size 4 - 1/2 inch spindle capacity

Style B - Three dimensional

Size 1 - 5/16 inch spindle capacity
Size 2 - 3/8 inch spindle capacity

| Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. |

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Interference and Susceptibility Requirements for the control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available the from Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2. Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues and Cutters. (DOD Adopted)

ANSI B5.10 - Machine Tapers. (DOD Adopted)

ANSI B11.8 - Drilling, Milling, and Boring Machines - Safety Requirements for the Construction, Care and Use. (DOD Adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept.,1430 Broadway, New York, NY 10018-3363.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

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(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The engraving machine shall be new and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and accessories necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the engraving machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be easily read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided that such materials will not jeopardize the intended use, performance, or design life of the engraving machine. Reclaimed materials shall have been collected or recovered from solid waste

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and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.2.3 Energy efficiency. The engraving machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the engraving machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.8. If a conflict arises between 29 CFR 1910 and ANSI B11.8, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The engraving machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the engraving machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine and its components, materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. An automatic lubrication system shall be provided for all ways and axes screws. After completion of axis motion, a low lube condition shall inhibit additional axes motion until proper lubrication is restored. The lube reservoir and pump for this system shall be self-contained, easily accessible, adjustable, and have adequate capacity to sustain the unit for a minimum of 24 hours of continuous operating without replenishing the reservoir.

3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replaceable parts except for hand-fitted parts such as ways shall be interchangeable without requiring modification. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign sourced

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replacement parts shall be interchangeable with domestically supplied components or be logistically supportable and readily available from a domestic source. Replacement parts are those that are normally subject to wear such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include motors, relays, switches, magnetics, pushbuttons, light bulbs, fuses, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall be completely interchangeable without need for any modifications.

3.3 Construction. The engraving machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent changes in tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the engraving machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the engraver and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 360.03a, AGMA 2000-A88 and AGMA 2001-B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461.

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The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. Type I machines shall consist of a base/table, column, spindle, pantograph assembly, copy holder base, and drive assembly. Type II machines shall consist of a column and base, knee, saddle, table, spindle, pantograph assembly, copy holder, and drive assembly. All ways shall be machined and scraped or shall be precision ground.

3.4.1 Base/table. The base/table of each type I machine shall be a close grain cast iron casting. The top of the base shall be machined flat and shall be used for mounting the workpiece.

3.4.2 Column. The column of each type I machine shall be a close grain cast iron casting. The column shall carry the pantograph assembly and copy table. A handwheel with a dial graduated in not greater than 0.001 inch increments shall be provided for vertical movement of the pantograph assembly and the copy table.

3.4.3 Column and base. The column and base of each type II machine shall be of close grain cast iron. The column shall be a box type casting, internally braced and ribbed to minimize distortion and deflection. The column ways shall be fitted to and support the knee and table with adjustable gibs or other means of adjustment for wear. The base shall be provided with not less than three mounting pads and means for securing the machine to the floor.

3.4.4 Knee. The knee of each type II machine shall be of a close grain cast iron. The knee shall house the gearing mechanism for vertical feed of the knee and cross feed of the table. A handwheel or crank and feed screw with dial shall be provided for actuating the above gearing mechanisms. The knee elevating screw shall be totally enclosed. Stationary covers or flaps shall be provided to protect the cross feed screw. Means shall be provided for locking the knee in position.

3.4.5 Saddle. The saddle of each type II machine shall be of close grain cast iron and shall be equipped with adjustable gibs for fitting to the ways of the knee. Means shall be provided for locking the saddle in position.

3.4.6 Table. The table of each type II machine shall be of close grain cast iron and shall be equipped with adjustable gibs for fitting to the ways of the saddle. The front face of the table shall be machined to be perpendicular to the top and parallel to the ways. The table shall be provided with not less than three equally spaced T-slots conforming to ANSI B5.1. A handwheel or crank and feed screw with dial shall be provided for the longitudinal travel of the table. Means shall be provided for locking the table in position.

3.4.7 Spindle. Each spindle shall be of alloy steel, hardened and precision ground, and shall revolve on pre-loaded ball bearings. Means for depth control shall be provided for vertical feed of the spindle and means shall be provided to lock the spindle in any position within its travel. The floating vertical movement shall be not less than 1/2-inch, with forming

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guide, on curved work. The spindle taper socket shall be in accordance with ANSI B5.10 and shall be capable of holding collets for straight shank cutters to the full capacity of the machine.

3.4.8 Pantograph assembly. The machine shall be equipped with a complete pantograph assembly. The pantograph bars shall be calibrated in such manner as to provide good visibility to the operator, and enable rapid, accurate setting of the slider blocks. Each flat bearing surface shall be machined to a finish not rougher than 32 microinches, and shall be flat to within 0.001 inch over its entire length. All joints in the pantograph assembly shall be fitted with replaceable ball or roller bearings. Bearings shall be of the pre-loaded type, or shall be provided with means for adjustment to compensate for wear.

3.4.9 Copy holder base. The copy holder base shall be designed to accept optional styles of V-groove, flat, or circular types and sizes of copy holders. Means shall be provided for accurate positioning and locking of the holder base in the traverse and longitudinal positions.

3.4.10 Drive assembly. All pulleys shall be dynamically balanced. Means for adjusting belt tension shall be provided.

3.4.11 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.12 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.13 Electrical system. The electrical system shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be wired to draw all of its electrical power from a single 230/460 volts, 3-phase, 60 hertz (Hz) circuit. Unless otherwise specified (see 6.2), the machine shall be initially wired for operation on 460 volts.

3.4.13.1 Motors. Motors shall meet the requirements of 3.4.13. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (ac) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.13.2 Worklight. Unless otherwise specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off

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switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.13.3 Hour meter. When specified (see 6.2), the engraving machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the non-resetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the engraving machine shall meet the requirements of table I for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each engraving machine shall meet the alignment tolerances specified in table II.

3.7 Performance. The engraving machine shall be capable of meeting performance requirements when tested in accordance with paragraph 4.8.4.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described see (6.2).

3.8 Standard equipment. All standard equipment normally furnished with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described see (6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

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3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate or chart shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Servicing interval
Type of lubricant
Federal of military specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall provide the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE or military/Federal specification number)
Type of filter

3.12 Workmanship. Workmanship of the engraving machine and its accessories shall meet all requirements specified herein and shall be of a quality to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and test) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractors overall inspection program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products

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or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the first article and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, all tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on the engraving machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The engraving machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 degrees to 104 degrees Fahrenheit and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. The engraver shall be operated at no load for not less than 1-hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance

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test given in 4.8.4 through 4.8.4.1. Any machine adjustments required to comply with 3.6 or after the performance test shall be cause for rejection.

4.8.4 Performance test. A 6.000 ± 0.0005 inch square master having corner angles of 90 degrees ± 30 minutes shall be duplicated onto a piece of commercial brass 1/4 inch thick, using a 3:1 reduction. The length of the sides and the corner angles of the brass workpiece shall be measured. The length of the sides shall be 2.000 ± 0.0025 inches and the corner angles shall be 90 degrees ± 30 minutes. In addition, 3-dimensional machines shall reproduce a master step block using 3:1 reduction. The step shall be 0.2500 ± 0.0005 high and 0.5000 ± 0.0010 wide. Height of finished workpiece step shall be 0.0830 ± 0.0020, with a width of 0.167 ± 0.0030.

4.8.4.1 Additional performance tests. Performance test required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2), level A preservation, level A r level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The machines covered by this specification are intended for use in toolrooms, machine shops, and laboratories.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, style, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

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g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

j. Electrical system, if different (see 3.4.13).

k. Initial voltage, if different (see 3.4.13).

l. Motors, if different (see 3.4.13.1).

m. Worklight, if not required (see 3.4.13.2).

n. Hour meter, if required (see 3.4.13.3).

o. Size and capacity, if different (3.5).

p. Specify physical size limitation, if required (see 3.5.1).

q. Additional performance requirements, if required (see 3.7.1).

r. Specify and fully describe optional equipment, if required (see 3.9).

s. Specify applicable repair parts, if required (see 3.10).

t. Lubrication plate, if required (see 3.11.1).

u. Information to be provided on lubrication plate, if different (see 3.11.1).

v. Nameplate, if different (see 3.11.2).

w. Hydraulic plate, if required (see 3.11.3).

x. Hydraulic plate, if different (see 3.11.3).

y. Inspection conditions, if different (see 4.3).

z. First article inspection, if different (see 4.4).

aa. Quality conformance inspection, if different (see 4.5).

bb. Acceptance test, if different (see 4.6).

cc. Additional performance test, if required (see 4.8.4.1).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language. Technical data shall be an exact duplication of the part, component, or system described.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a production sample, a first article sample, a first production item, a sample selected from the first __ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in Inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

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6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

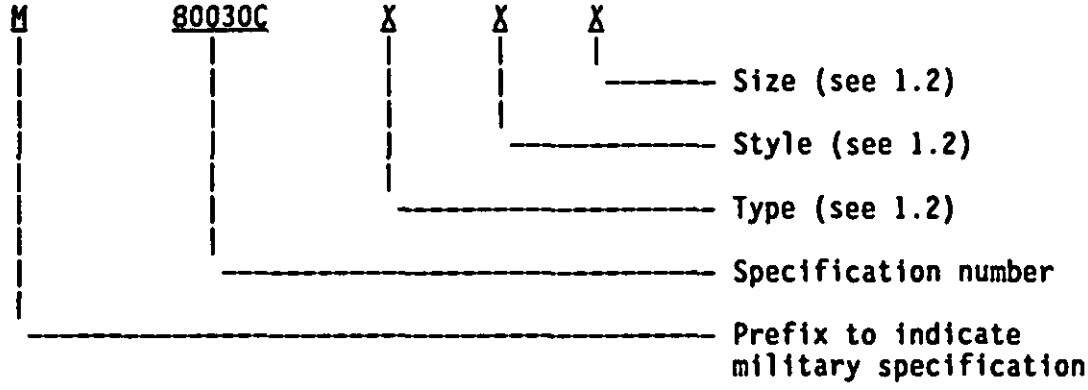
6.8 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.13 Subject term (keyword) listing.

Bench
Column
Pantograph
Profile

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TABLE I. Size and Capacity

CHARACTERISTIC	REQUIREMENT									
	Type I, Bench Style A				Type II, Floor Style A				Type II, Floor Style B	
	Size 1	Size 2	Size 3	Size 4	Size 1	Size 2	Size 3	Size 4	Size 1	Size 2
Spindle:										
Capacity	1/8"	1/4"	1/4"	5/16"	1/4"	5/16"	3/8"	1/2"	5/16"	3/8"
Speeds, No. of	1	2	6	4	6	8	9	12	8	9
Speed range, rpm	14,000 (1 spd)	14,000 18,000	4,500- 26,000	5,800- 18,000	3,500- 20,000	4,700- 8,000	1,300- 11,500	900- 12,500	4,700- 18,000	1,300- 13,000
Vertical feed	3/8"	21/32"	5/16"	5/16"	1/4"	5/16"	7/16"	1"	5/16"	1"
Maximum spindle nose to work table, at least	-	21/32"	10"	10"	6"	11"	12"	16"	10"	13"
Vertical adjustment of Spindle assy.	3/8"	21/32"	10	10	-	-	-	-	-	-
Engraving area, one setting	2 x 9	6 x 18	5 x 13	6 x 19	-	-	-	-	-	-
Maximum circle diameter one setting	3-3/8"	7"	7"	10"	-	-	-	-	-	-
Copy table size	-	10 x 24	10 x 24	10 x 24	-	-	-	-	-	-
Work table:										
Work surface size	-	-	-	-	6"x 12"	8"x 18"	9"x 26"	10"x 42"	8"x 14"	9"x 34"
T-slots, No. and size	-	-	-	-	3, 13/32"	4, 7/16"	3, 1/2"	3, 11/16"	4, 7/16"	3, 1/2"
Longitudinal travel	-	-	-	-	7"	10"	16"	22"	6"	22"
Cross travel	-	-	-	-	7"	8"	8"	10"	8"	16"
Vertical travel	-	-	-	-	6"	11"	12"	14"	10"	13"
Optional compound work table size:										
Longitudinal travel	5-1/2"x8"	-	9"x 18"	8"x 17"	-		-	-	-	-
Cross travel	2"	-	6"	9"	-		-	-	-	-
Reduction ratio range	2-1/2:1 to 6:1	2:1 to 8:1	2:1 to 32:1	1:1 to 100:1	1:1 to 100:1	1:1 to 40:1	1:1 to 40:1	1:1 to 6:1	2:1 to 10:1	2:1 to 8:1
Motor horsepower	1/15	1/15	1/6	1/5	1/3	1/4	1/2	1	1/4	1/2

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TABLE II Alignment accuracy.

DESCRIPTION	ALLOWABLE DEVIATION NOT MORE THAN
1. Table square with column slide	0.002 inch in 12 inches
2. Cutter head traverse parallel with table	0.005 maximum in 8 inches
3. Spindle square with table (sweep with 4" radius)	0.004 inch TIR
4. Spindle runout	0.0005 inch 1 inch from nose
5. Tracer traverse parallel with copy holder	0.015 inch in 12 inches
6. Tracer sweep parallel with copy holder	0.005 inch in 1 inch

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CONCLUDING MATERIAL

Custodians:
Army - AL
Navy - SH
Air Force – 84

Preparing Activity:
DLA - IP

(Project 3417-0187)

Review Activities:
Navy - SH

User Activities:
Navy – OS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-80030C	Milling Machine, Two and Three Dimensional Engraving

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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